

Samuel Monteiro Junior

FULL-STACK ENGINEER

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PROFESSIONAL PROFILE

Full-stack engineer who ships the entire path: from the Next.js and React interface users touch to the FastAPI and Node services and the PostgreSQL schema underneath. Comfortable across the stack: typed REST APIs, real time over WebSockets, background jobs with BullMQ, and the data modeling that keeps everything fast under load. What I build reaches production with its own reliable infrastructure, without handing it off to another team.

TECHNICAL SKILLS

Frontend: React, Next.js, TypeScript, Vite, Tailwind CSS, Zustand, TanStack Query, Zod, Shadcn UI, Three.js

Development & Automation: Python (FastAPI, Web Scraping, CLI Scripts), Node.js (REST APIs, WebSockets, BullMQ, Redis), Rust (Tokio, Axum, Hyper), n8n Workflows, Shell/Bash Scripting for operational task automation

Databases & Cryptography: PostgreSQL, TimescaleDB, SQL Server, Redis Pub/Sub, SQLite, LUKS Full Disk Encryption, mTLS

PROFESSIONAL EXPERIENCE

Automation Engineer & Full Stack Developer · Freelancer

April 2022 to Present

- Architected and deployed microservices ecosystems using FastAPI and Node.js in Docker containers, reducing request processing times by 40%.
- Designed and implemented a private mesh network (Tailscale VPN) with mTLS authentication to connect local hosts to Debian VPS servers, securing sensitive data traffic.
- Developed Cortex-Nexus, an automation pipeline (n8n, Python, AI) for lead extraction and qualification, achieving 99.7% deliverability and processing 3,000+ leads per month.
- Structured Parana-Leads, with automated deployment in Docker containers using Caddy as a reverse proxy with automatic SSL, reducing deployment time for new environments by 70%.

Support Analyst / Database Administrator · CNI Tecnologia e Software

June 2024 to December 2024

- Administered Linux/Windows network security policies and servers, reducing security incidents by 25% through connection hardening and access control.
- Developed Bash and Python scripts to automate relational database backups and hardware performance telemetry, reducing manual monitoring effort by 15 hours per week.
- Optimized queries and maintenance routines in SQL Server and PostgreSQL databases, reducing management report load times by 50%.

Support Analyst & Web Developer · Vip Informática

February 2017 to January 2021

- Administered operational databases (PostgreSQL/SQL Server), implementing automatic backup routines and reducing downtime to near zero.
- Developed and maintained internal tools (Node.js/PHP) focused on query and load time optimization, accelerating internal API response times by 35%.
- Led the restructuring of IT support processes and technical documentation, resulting in a 30% reduction in repetitive support tickets.

PROJECTS

TessTrade: Algorithmic Trading Platform

[Rust](#) / [React](#) / [Python](#) / [TimescaleDB](#)

Live platform at tesstrade.com combining social features, a quantitative development environment, and backtesting: multithreaded Rust simulation engine, sandboxed Python indicators connected to the engine via FFI/RPC, and a React workspace with Monaco Editor for authoring strategies in the browser.

6ID: License Ecosystem for Trading Platforms

[FastAPI](#) / [Next.js](#) / [C# WPF](#) / [PostgreSQL](#)

Distributed ecosystem for license management and user authentication on trading platforms: asynchronous FastAPI backend (SQLAlchemy, Asyncpg), Next.js administration console with validation mirrored between Pydantic and Zod, and a Windows WPF client on .NET 8, compiled with ReadyToRun into a single binary.

BoxOptimizer API

[.NET 8](#) / [JWT](#) / [Docker](#)

High-performance API for three-dimensional logistics calculation (efficient box cubing) using the First Fit Decreasing (FFD) algorithm, packaged as a Docker image and ready for horizontal scaling.

EDUCATION

Associate's Degree in Systems Analysis and Development

2024, ongoing

PUCPR (Distance Learning)

Bachelor of Laws (LL.B.)

2016 to 2021

UNISEPE

CC50: Introduction to Computer Science

2023

Harvard University / Fundação Estudar